1. Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) is the most basic level of cloud computing. In this model, the cloud provider offers virtualized computing resources over the internet, such as virtual machines, storage, networking, and servers.

Instead of buying physical hardware, organizations rent infrastructure on a pay-as-you-go basis.

In IaaS, the cloud provider manages the physical hardware, including data centers, servers, networking, and virtualization.

However, the user is responsible for managing the operating system, middleware, runtime, applications, and data.

This gives users high control and flexibility, similar to traditional on-premise systems but without the cost of maintaining hardware.

IaaS is mainly used by system administrators and IT teams who want complete control over their infrastructure.

It is ideal for workloads that require customization, such as hosting applications, running virtual machines, disaster recovery, and large-scale data processing.

One of the biggest advantages of IaaS is scalability. Resources can be increased or decreased instantly based on demand.

For example, in Azure, you can increase the number of virtual machines just by changing a number in a configuration or textbox, making it highly efficient compared to manual server setup.

IaaS – Real-Time Example

Imagine you are running an e-commerce website (like a small Amazon clone).

Instead of buying physical servers, you use Azure Virtual Machines (VMs):

- You create 5 virtual servers to host your website.
- During a sale (Diwali/Black Friday), traffic increases.
- You simply increase the number of VMs from 5 to 20 using Azure (auto-scale).
- After the sale, you reduce it back to 5 to save cost.

Here, you are managing:

- OS installation (Linux/Windows)
- Application setup (Tomcat, Node.js, etc.)
- Database installation

Real tools used: Azure VM, AWS EC2

2. Platform as a Service (PaaS)

Platform as a Service (PaaS) provides a complete development and deployment environment in the cloud.

It includes not only infrastructure but also operating systems, development tools, databases, middleware, and runtime environments.

This allows developers to focus purely on writing code without worrying about managing servers or system configurations.

In PaaS, the cloud provider manages infrastructure + OS + runtime, while the user is responsible only for application development and data.

This significantly reduces development complexity and speeds up the software development lifecycle.

PaaS is mainly used by developers who want to build, test, and deploy applications quickly.

It supports multiple programming languages and frameworks, and often includes built-in services like authentication, scaling, and monitoring.

A major advantage of PaaS is rapid development and deployment. Teams can collaborate easily, and applications can be launched faster without infrastructure setup. However, it provides less control compared to IaaS.

PaaS – Real-Time Example

Now suppose you are a developer building a web application.

You use Azure App Service:

- You write your code (Java, Python, Node.js)
- Upload it directly to Azure
- No need to install OS, server, or runtime
- Azure automatically handles:
    - Deployment
    - Scaling
    - Load balancing

Example:

- You build a college portal or banking app
- Deploy it in one click
- If users increase, Azure auto-scales

Real tools used: Azure App Service, Google App Engine

3. Software as a Service (SaaS)

Software as a Service (SaaS) is the highest level of cloud computing where complete software applications are delivered over the internet. Users do not need to install, manage, or maintain anything.

They simply access the application through a web browser.

In SaaS, the cloud provider manages everything: infrastructure, platform, application, updates, security, and maintenance.

The user only interacts with the software and data.

SaaS is designed for end-users and businesses that want ready-to-use applications without technical complexity.

It is widely used for daily business operations such as email, document management, CRM, and collaboration tools.

The biggest advantage of SaaS is ease of use and zero maintenance. Users can access applications from anywhere, and updates happen automatically.

However, customization and control are limited compared to IaaS and PaaS.

SaaS – Real-Time Example

Think about tools you already use daily:

Example 1: Gmail

- You just open browser and send emails
- No need to install server or manage storage

Example 2: Microsoft Teams / Zoom

- Join meetings directly
- No backend setup needed

Example 3: Google Docs

- Create and share documents online
- No software installation required